

AOMORI, Japan

WMO: 475750

Lat: 40.8219N

Lon: 140.7689E

Elev: 13

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.2	(c) 22.0	(d) 13.7	(e) 11.0	(f) 26.3	(g) 15.5	(h) 12.1	(i) 27.4	(j) 25.1	(k) 30.5	(l) 22.8	(m) 30.0	(n) 10.4	(o) 210	(p) 0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.3	(c) 87.1	(d) 75.1	(e) 84.2	(f) 73.8	(g) 81.6	(h) 72.7	(i) 76.8	(j) 84.3	(k) 75.5	(l) 82.1	(m) 74.3	(n) 80.1	(o) 12.4	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 74.6	(b) 129.5	(c) 81.0	(d) 73.4	(e) 124.4	(f) 79.8	(g) 72.2	(h) 119.2	(i) 78.9	(j) 40.3	(k) 84.4	(l) 39.1	(m) 82.4	(n) 37.9	(o) 80.2	(p) 81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.3	(b) 20.6	(c) 18.3	(d) DB	(e) 16.5	(f) 92.4	(g) 2.4	(h) 3.2	(i) 14.8	(j) 94.7	(k) 13.4	(l) 96.5	(m) 12.0	(n) 98.3	(o) 10.3	(p) 100.7	(q) 82.7
			(d) WB	(e) 15.5	(f) 78.5	(g) 2.3	(h) 1.6	(i) 13.9	(j) 79.7	(k) 12.5	(l) 80.6	(m) 11.3	(n) 81.5	(o) 9.6	(p) 82.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	30.1	31.2	37.0	47.7	57.2	64.1	72.0	75.0	68.2	56.7	44.9	34.6
	DBStd	16.40	4.31	4.89	5.61	6.04	5.13	4.53	5.06	4.58	5.33	5.35	6.78	5.40
	HDD50	2331	618	527	404	113	5	0	0	0	0	8	178	478
	HDD65	5571	1083	947	867	521	248	69	4	0	25	262	603	942
	CDD50	2939	0	0	2	43	228	423	683	774	546	215	25	0
	CDD65	705	0	0	0	1	6	42	222	309	121	4	0	0
	CDH74	3934	0	0	0	7	47	150	1175	2004	544	7	0	0
	CDH80	917	0	0	0	0	6	23	278	506	104	0	0	0
Wind	WSAvg	8.3	9.3	9.4	9.8	9.4	8.2	7.2	6.9	6.8	6.8	7.6	8.6	9.2
Precipitation	PrecAvg	53.5	5.7	4.0	3.0	2.7	3.1	3.0	5.1	5.6	5.2	4.6	5.5	6.1
	PrecMax	65.6	10.9	7.4	7.1	5.6	6.7	6.5	10.4	12.5	13.6	9.5	12.5	9.4
	PrecMin	39.6	2.5	1.3	0.7	0.6	1.5	0.4	0.8	1.0	1.5	1.7	2.5	3.7
	PrecStd	8.0	1.8	1.6	1.4	1.2	1.3	1.5	2.6	3.2	2.7	2.2	2.1	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	49.0	60.6	73.7	79.5	83.6	90.2	92.0	87.6	74.4	66.5	54.1
		MCWB	39.3	42.4	48.5	56.0	60.2	67.3	75.8	76.7	75.6	63.2	55.9	48.7
	2%	DB	40.7	43.9	53.8	66.3	73.6	78.0	86.3	88.0	82.9	71.2	62.1	49.4
		MCWB	36.6	38.3	45.0	51.9	58.4	66.0	74.4	76.0	72.6	61.7	54.2	44.5
	5%	DB	38.7	41.3	49.7	61.7	69.7	74.7	82.9	85.0	79.4	68.8	58.8	45.9
		MCWB	35.0	36.5	42.7	49.8	56.8	64.7	73.5	75.0	70.4	60.3	52.1	41.5
	10%	DB	36.7	38.9	46.1	57.8	66.5	71.9	80.1	82.6	76.8	66.3	55.6	42.8
		MCWB	33.4	34.8	40.0	48.3	56.4	63.7	72.1	73.9	68.6	58.6	50.3	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	43.3	49.9	58.0	63.7	70.5	77.3	78.8	76.9	67.4	58.5	49.9
		MCDB	43.1	47.7	57.5	67.1	72.6	77.7	86.1	86.7	84.1	71.6	62.9	53.4
	2%	WB	37.4	39.3	46.2	54.4	61.2	68.1	75.9	77.5	74.4	63.9	55.4	45.4
		MCDB	40.2	42.8	52.8	62.5	69.2	74.6	83.6	84.8	80.7	68.6	60.4	48.7
	5%	WB	35.5	37.1	43.1	51.8	59.4	66.4	74.6	76.4	72.3	61.9	53.2	42.1
		MCDB	38.0	40.5	48.5	58.7	66.8	72.5	81.3	83.1	77.6	66.9	57.7	45.4
	10%	WB	33.8	35.0	40.8	49.7	57.7	64.9	73.3	75.2	70.2	59.8	50.8	39.1
		MCDB	36.2	38.3	45.5	56.7	64.5	70.3	78.8	81.0	75.3	64.9	55.0	42.4
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.8	12.5	15.7	15.2	12.9	11.8	12.3	14.3	15.2	12.9	9.5
		MCDBR	11.7	14.0	18.6	22.0	21.4	17.4	15.6	15.4	16.5	18.0	18.6	15.1
		MCWBR	9.2	10.5	12.4	12.2	10.6	8.5	6.8	7.1	8.4	10.9	12.6	11.7
	5% WB	MCDBR	11.8	14.2	17.7	18.5	17.7	14.2	13.4	13.2	14.6	15.0	17.6	15.2
		MCWBR	10.0	11.4	12.9	12.0	10.0	7.6	6.3	6.5	8.3	10.4	13.4	12.5
Clear-Sky Solar Irradiance	taub	0.377	0.421	0.493	0.542	0.556	0.531	0.534	0.500	0.441	0.420	0.408	0.391	
	taud	2.006	1.899	1.826	1.784	1.831	1.972	2.034	2.140	2.259	2.183	2.108	2.038	
	Ebn at Noon	231	239	236	235	236	242	240	245	252	240	220	214	
	Edh at Noon	38	50	60	67	66	57	53	47	39	37	35	35	
All-Sky Solar Radiation	RadAvg	410	645	1017	1404	1575	1575	1385	1367	1172	878	526	362	
	RadStd	27	67	84	144	158	131	144	139	92	66	48	30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page